GPS MAP MATCHING USING HMM AND VITERBI

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- **Introduction**

  GPS trajectories often contain noise and may not align exactly with real roads. Map matching addresses this by associating GPS observations with the most likely road segments in a digital road network. This project implements an HMM-based map-matching pipeline using GeoLife GPS trajectories and the Beijing road network from OpenStreetMap (OSM).

- **Methodology**

  The workflow loads a GPS trajectory and a directed OSM road graph projected into metric coordinates. For each GPS observation, nearby road segments are generated as candidates. The Hidden Markov Model (HMM) evaluates these candidates using emission and transition probabilities. Emission probability measures how well a road candidate fits an observed GPS point, primarily based on GPS-to-road distance. Transition probability evaluates whether movement between consecutive candidate roads is plausible by comparing observed GPS movement with road-network distance and connectivity.

- **Role of Viterbi**

  After calculating the HMM scores, the Viterbi algorithm finds the highest-scoring sequence of road candidates across the trajectory. Rather than independently choosing the nearest road for every GPS point, it considers the sequence as a whole and balances point-to-road fit with movement consistency. The selected road sequence is then visualized alongside the original GPS observations.

- **Results**

  For the evaluated sample, 908 GPS observations were processed. Road candidates were available for 879 observations, giving 96.8% candidate coverage. The Viterbi output contained 559 matched observations. These figures describe candidate availability and the generated output; they do not establish ground-truth matching accuracy. A map visualization is also produced to compare the original GPS points with the predicted route.

- **Conclusion and Future Work**

  The project demonstrates an end-to-end map-matching pipeline combining GeoLife GPS data, an OSM road graph, HMM emission and transition scoring, and Viterbi decoding. Results can be affected by GPS noise, missing nearby candidates, and disconnected road segments. Future work should compare predicted routes with ground-truth data, calculate suitable accuracy metrics, and measure runtime under controlled conditions. As the current pipeline is offline, a real-time delay-versus-accuracy trade-off has not yet been evaluated.